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655.662 — Introduction to Healthcare Systems Engineering
Module 6 | M6A3: Team Learning

How can “Team Learning” enable your remote care system?

Senge defines “Team Learning” as the process of achieving alignment (not necessarily agreement) among the members of a team to develop the collective ability of the team to achieve the desired outcomes for all team members. As such, when a team is out of alignment, they waste energy: each member works hard at something but does not contribute to the success of the entire team. This represents a situation where individual efforts are like a light bulb radiating in multiple directions while a well-aligned team produces a laser-like effect. Senge identifies the three dimensions required for successful Team Learning as follows:

1. Developing insightful thinking regarding complex problems. Teams need to collectively think more intelligently than any one person.
2. Coordinated action through innovation. Rather than relying on instruction, trust-based operational processes are used.
3. Obligations of a learning team include disseminating knowledge and best practices to additional teams.

Why PD-RPM requires Team Learning more than most other programs.

The teams involved in PD-RPM do not share common physical or employment locations. Therefore, there are several different teams including those located at the dialysis facilities where patients receive treatment (home dialysis), engineering teams responsible for monitoring centers, the company providing cyclers (the equipment that performs the dialysis procedure), teams within carriers’ network operation departments, companies providing integration support for electronic health records, and patients and caregivers who manage the system daily. Most of these teams cannot be directed by the project due to the fact that PD-RPM is an acknowledged system of systems. Additionally, failure is a cross-boundary issue by design. For example, the “silent-gateway risk” listed in the risk register cannot be identified solely based upon data collected by a single team. Data associated with this risk includes logs maintained by the carrier’s network operations department, information contained in firmware related to the gateway device, information placed in queues by nurses, and conditions in patients’ homes. No one team maintains sufficient amounts of this data.

Seven practices to establish it. Team Learning is a discipline, not simply a personality trait. Therefore, it must be embedded into how the program conducts itself.

Collaborative practice	Chapter 11 principle it implements	Who takes part	What it produces
Weekly silent-patient review	Dialogue: suspend assumptions — no cause may be named in the first 10 minutes	Nurse, firmware lead, carrier engineer	A shared account of what happened, before anyone defends a position
Rank left at the door	Dialogue: regard one another as colleagues	All participants	The nurse’s observation weighs the same as the architect’s
A named facilitator who does not decide	Dialogue: someone must hold the context	Systems engineer	The conversation stays exploratory long enough to be useful
Converge-and-assign close	Discussion: the team moves deliberately to decision	Same group	One change, one owner, one date
Shared timestamped evidence log	Surfaces defensive routines by removing ownership of the facts	All parties, including the carrier	“Whose fault” becomes “what happened”
Patients and care partners seated in the review	Dimension one: many minds more intelligent than one	Patient, care partner	Access to the half of the system no instrument observes
Fleet replay sandbox	The practice field — a microworld to rehearse in	Engineering and nursing together	Rehearsal of a failure without patient consequence

Senge describes two types of communication in a team environment. Discussion involves sharing ideas with others with the objective of finding the “winner.” While this method of communicating can result in a team reaching consensus quickly, it shares a root with “percussion,” where ideas are exchanged rapidly until one idea emerges as the winner. Dialogue, however, provides a free flowing exchange of meaning where participants observe their own thought patterns. Bohm’s three conditions provide the framework for creating dialogue: suspend assumptions, treat others as colleagues, and have someone maintain context. The first six practices previously identified establish Bohm’s three conditions. A learning team learns how to transition between discussion and dialogue. Defensive routines (i.e., habits designed to prevent embarrassment and preserve mediocrity) limit our ability to engage in dialogue. According to Argyris, defensive routines are predictable and attribution is the primary routine limiting PD-RPM’s ability to engage in dialogue. When the carrier states that the problem lies with the gateway, the firmware lead responds that the problem must lie with the carrier, and the nurse states that she believes that the system must have been down once again. All parties are protecting themselves from being determined as the source of the problem and the shared evidence log is intended to eliminate the need for this protection.

A scenario: phantom Sunday outage.

During the clinical pilot phase of the program, 11 patients logged silent-device events that were concentrated on Sunday evenings over a period of 3 consecutive weeks. The carrier reviewed its logs and stated that there was no outage recorded by its network operations department. Firmware telemetry indicated that there were no faults detected by the firmware related to the gateway device and a full battery history existed for the gateway battery. The nurse reported that patients stated that they experienced no problems with respect to using their device. Each team examined its own evidence and concluded that the problem lay with another team. The program almost wrote off the observed events as instrumentation-related noise -- the typical defensive response resulting from three separate analytical evaluations that add up to no learning. At this point, a dialogue session was scheduled for all three teams plus two patients participating in the pilot testing, with a rule requiring that no cause be named within 10 minutes of starting. After 9 minutes of dialogue, a patient provided information indicating that they turned off their power strip prior to going to sleep each Sunday evening because of visits from housekeepers during this timeframe. The gateway battery has a 6-hour life and Sunday night sessions typically continue longer than 6 hours. None of the participants would have individually discovered this information: the carrier had documented no outage, firmware had documented no errors, nurses had received no complaints from patients, and patients were unaware that turning off their power strip was relevant to maintaining proper function of their system. As a result, a 24-hour battery was added to future versions of the system, language was added to patient education materials instructing them to never turn off their power strips during use and all subsequent technique failure reviews included seating patients along with representatives from all applicable teams.

The last outcome illustrates Senge’s third dimension at work: technique failure reviews conducted by home dialysis programs now employ the same format as those performed by PD-RPM and therefore this approach is spreading beyond PD-RPM to other teams. Why this is classified as advanced development versus post-fielding is that this marks the first time that these teams have ever had to generate output collaboratively. If teams fail to develop their ability to communicate effectively before integrating into a functioning system, they will certainly fail to develop this capability after they begin working together.